TECHNICAL REPORT

Implementation and Evaluation of FastSpeech 2 for English and Telugu Text-to-Speech Synthesis

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Project: FastSpeech 2 Multilingual TTS Implementation

Languages: English and Telugu

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This report documents the implemented pipelines, experimental setup, training results, speech generation outputs, and limitations for two separate FastSpeech 2 text-to-speech systems.

Abstract

This project implements separate FastSpeech 2 text-to-speech (TTS) pipelines for English and Telugu. The systems use language-specific text processing, acoustic feature preparation, duration supervision, pitch and energy features, model training, and inference. Five test utterances were synthesized for each language. Duration, pitch, and energy visualizations were generated for every utterance. Griffin-Lim was used for waveform reconstruction from predicted mel spectrograms. The experiments demonstrate an end-to-end implementation, while the limited training subset and short training schedule are important constraints on the quality and generalization of the generated speech.

1. Introduction and Objectives

Text-to-speech synthesis converts written text into spoken audio. FastSpeech 2 is a non-autoregressive acoustic model that predicts mel-spectrogram representations from input text. Its variance adaptor models duration, pitch, and energy to provide explicit control over prosodic characteristics.

Project objectives

• Implement separate FastSpeech 2 pipelines for English and Telugu.

• Prepare language-specific text and acoustic features.

• Train and validate each model using separate data splits.

• Generate speech for five test utterances per language.

• Visualize predicted duration, pitch, and energy for each test utterance.

• Compare the experimental outcomes and document limitations.

2. FastSpeech 2 Architecture

The implemented architecture follows the FastSpeech 2 structure. A text encoder converts token sequences into hidden representations. The variance adaptor adds duration, pitch, and energy information. The decoder transforms the conditioned representations into mel spectrogram frames. A waveform reconstruction stage converts the predicted mel representation into audio.

Component	Function
Text encoder	Encodes input phoneme/character tokens into hidden representations.
Duration predictor	Predicts the number of acoustic frames associated with each input token.
Pitch predictor	Models fundamental-frequency-related information.
Energy predictor	Models frame-level acoustic energy.
Variance adaptor	Combines duration, pitch, and energy information with encoder outputs.
Mel decoder	Predicts the output mel spectrogram.
Vocoder / reconstruction	Converts the mel spectrogram into a waveform; Griffin-Lim was used here.

The two language implementations use separate model directories, configurations, preprocessing artifacts, checkpoints, and inference outputs.

3. Dataset and Experimental Setup

Item	English	Telugu
Dataset	Svarah	SPRINGLab IndicTTS Telugu

Item	English	Telugu
Source split	4,662 train / 518 validation records	199 usable aligned utterances
Source duration	7.18 h train / 0.79 h validation	Approximately 1.27 h usable audio
Prepared model subset	300 aligned samples	199 aligned samples
Training subset	270	179
Validation subset	30	20
Training epochs completed	1	1
Training loss	7.1523	8.5027
Validation loss	5.1646	5.2605
Compute	CPU	CPU

Dataset scope clarification: The English source split contains approximately 7.18 hours of training audio and 0.79 hours of validation audio, but only 300 aligned samples were prepared and used for the reported model run. The Telugu experiment used 199 usable aligned utterances, totaling approximately 1.27 hours. These experiments do not meet the originally specified five-hour training and one-hour validation target for each language.

4. English Pipeline

The English pipeline uses the Svarah dataset. Audio was processed into mono acoustic inputs and resampled to 22,050 Hz for feature preparation. The mel representation uses 80 mel channels, an FFT size of 1,024, a hop length of 256, and an 8,000 Hz upper mel-frequency limit.

Text normalization expands numbers and abbreviations, normalizes text to the model's supported uppercase symbol inventory, and represents word boundaries using the configured separator. A pretrained English wav2vec 2.0 CTC model was used to obtain alignments. These alignments provide duration supervision. WORLD-based F0 extraction and RMS energy features were prepared for the variance adaptor.

The English model was trained for one completed epoch using 270 training examples and evaluated on 30 validation examples. The resulting training loss was 7.1523 and validation loss was 5.1646.

English training curves

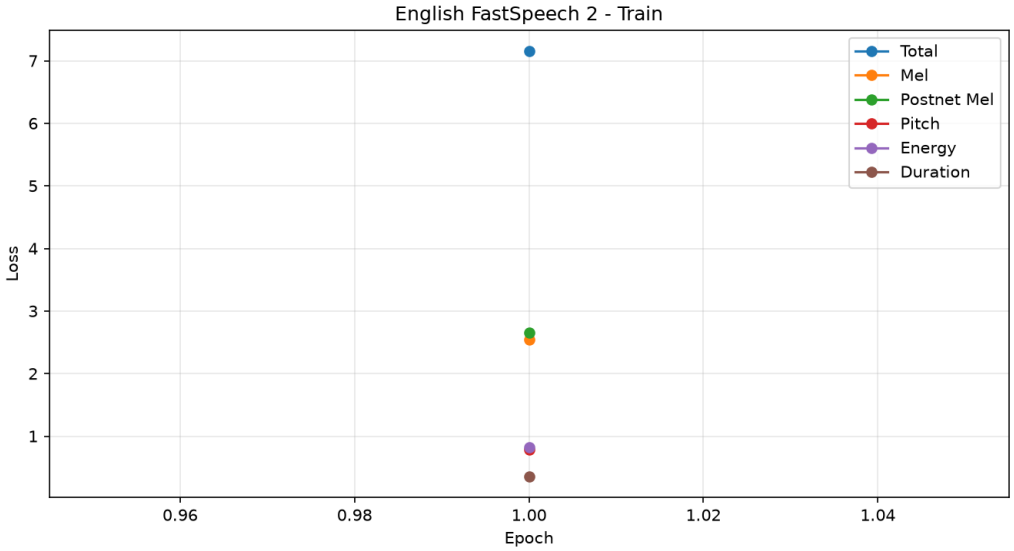


Figure 1. English training loss.

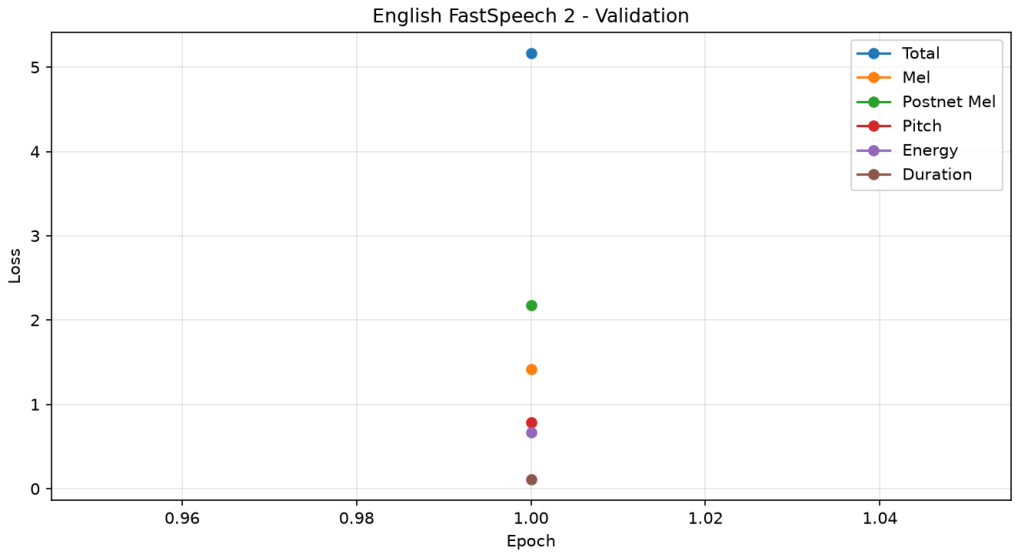


Figure 2. English validation loss.

5. Telugu Pipeline

The Telugu pipeline uses the SPRINGLab IndicTTS Telugu dataset. A Telugu wav2vec 2.0 CTC alignment model was used to obtain token alignment information. Audio was processed at 16,000 Hz with an 80-channel mel representation, FFT/window size 1,024, hop length 256, and a 7,600 Hz upper mel-frequency limit.

Telugu-specific preprocessing includes script-aware text handling, CTC alignment, character-level duration allocation, WORLD-based pitch extraction, and energy feature preparation. One utterance was excluded because it contained a character unsupported by the configured aligner vocabulary. The resulting split contained 179 training and 20 validation utterances.

The Telugu model completed one epoch with training loss 8.5027 and validation loss 5.2605.

Text normalization note: Some Telugu synthesis inputs were normalized to accommodate the available vocabulary and aligner. In particular, an unsupported character in T1 led to replacement of the original word with a simpler word; T3 spelling was corrected, and numerals in T4 were expanded into Telugu words. The original assignment text and the actual synthesis input are retained in inference metadata.

Telugu training curves

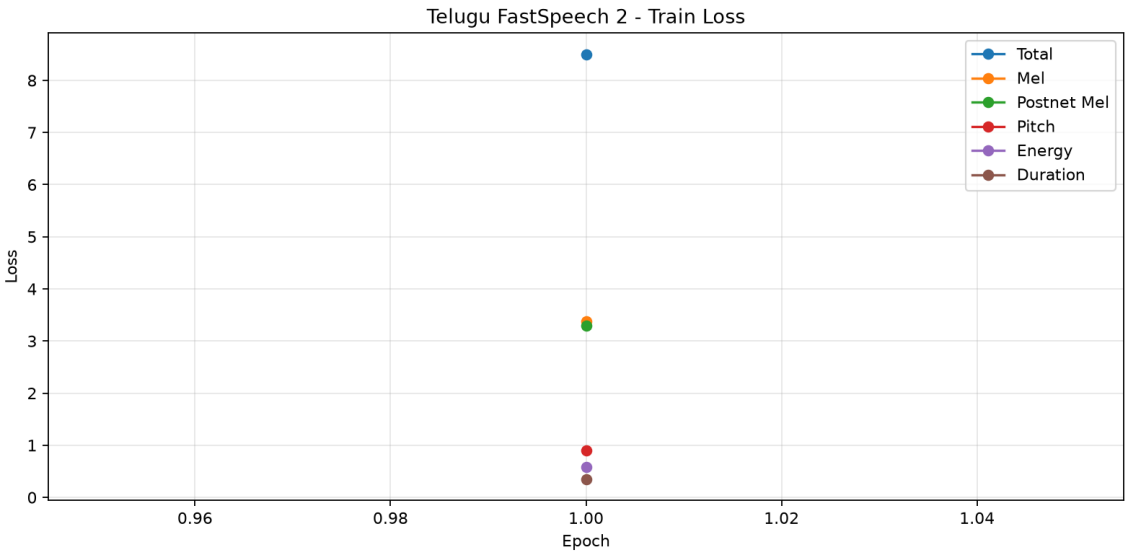


Figure 3. Telugu training loss.

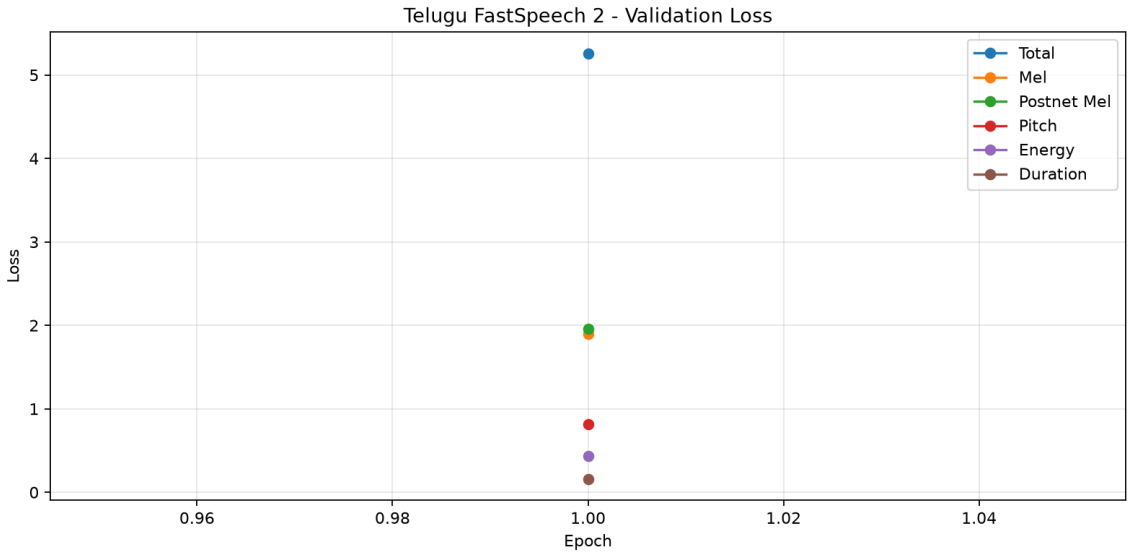


Figure 4. Telugu validation loss.

6. Speech Generation and Test Samples

Inference was performed using the best saved checkpoint for each language. Five utterances were generated for English (E1, E2a, E2b, E3, E4) and five for Telugu (T1, T2a, T2b, T3, T4). The generated waveforms were reconstructed from predicted mel spectrograms using the Griffin-Lim algorithm with 12 iterations.

Language	Sample IDs	Generated audio location
English	E1, E2a, E2b, E3, E4	English/outputs/english/test_generated_audios/
Telugu	T1, T2a, T2b, T3, T4	Telugu/outputs/telugu/test_generated_audios/

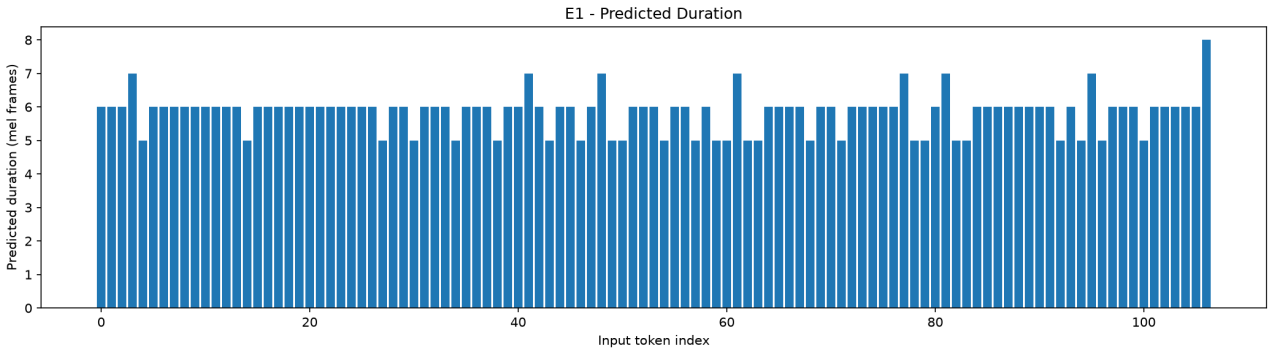
The English test set includes a long technical word, the two different uses of 'conduct', an intentionally unusual 'rhyth m' spelling, and a sentence containing a time, date, and proper name. The Telugu set contains the supplied test items, with the documented normalization changes described in Section 5.

7. Variance Adaptor Analysis

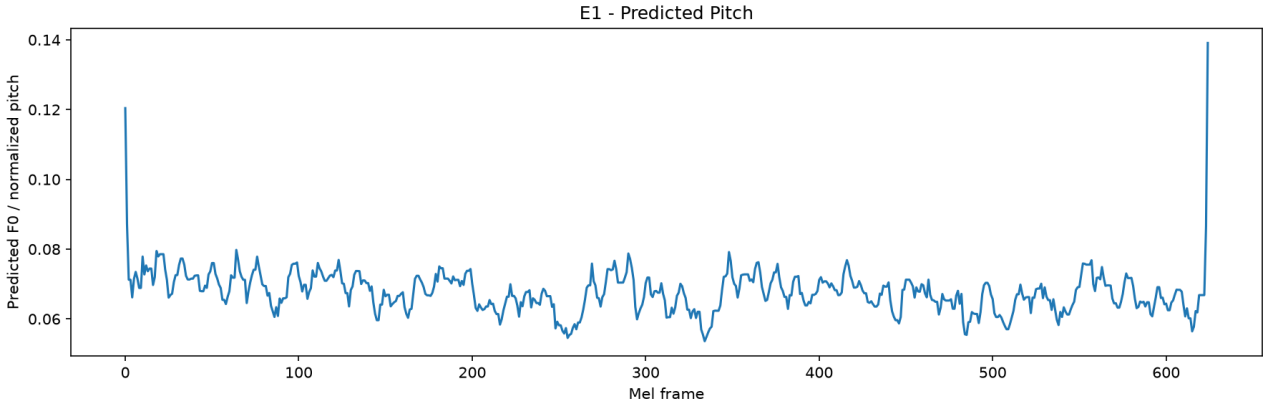
For every generated utterance, three plots were saved. Duration plots show predicted acoustic-frame allocation across input tokens. Pitch plots show the predicted pitch trajectory, and energy plots show the predicted energy trajectory over mel frames. These visualizations allow inspection of the model's predicted temporal and prosodic patterns. They are model predictions and should not be interpreted as objective measurements of naturalness or intelligibility.

English variance-adaptor plots

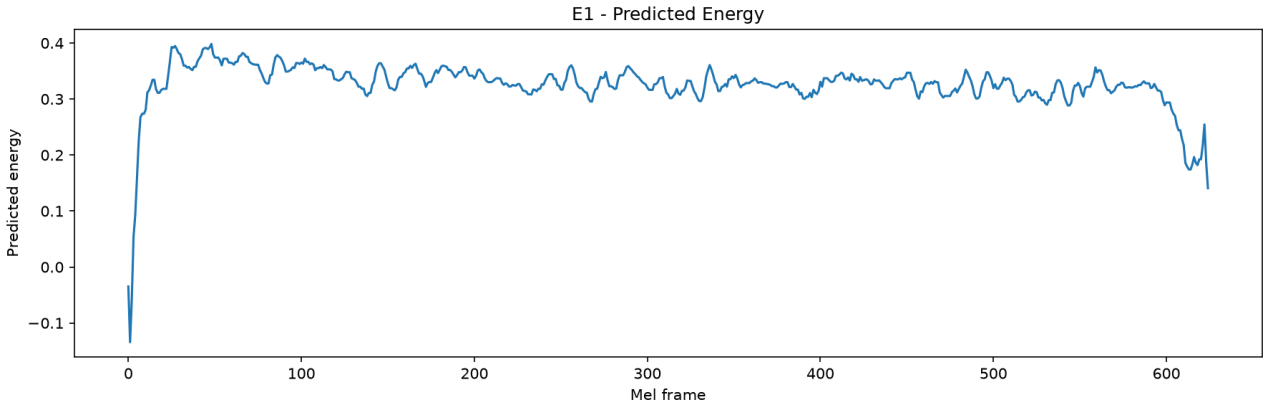
English - E1



E1: predicted duration.

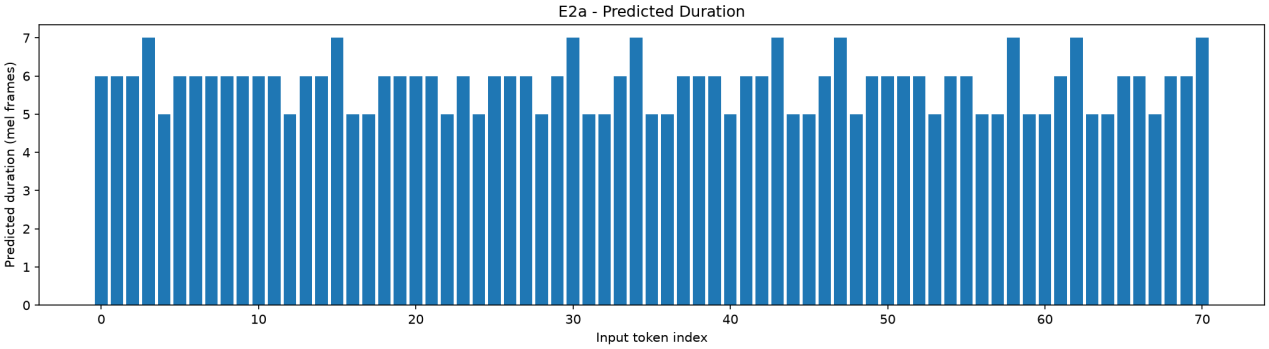


E1: predicted pitch.

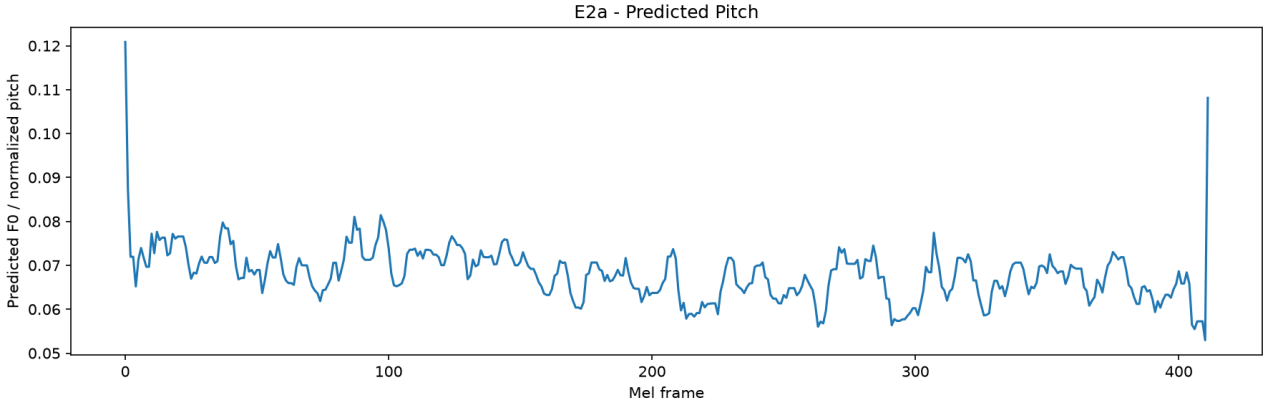


E1: predicted energy.

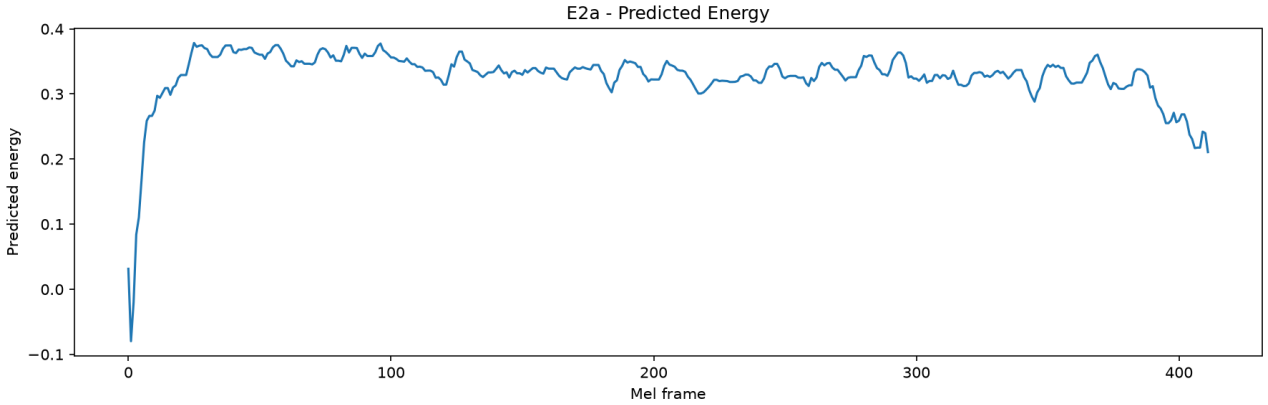
English - E2a



E2a: predicted duration.

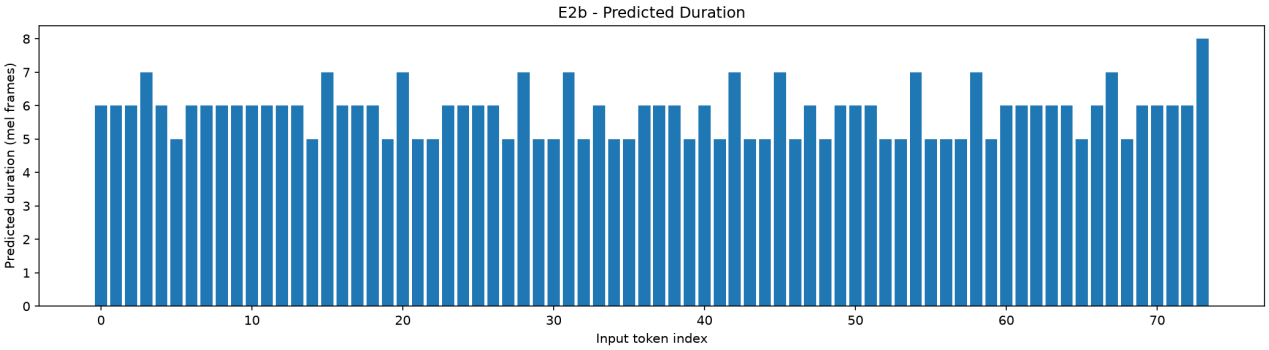


E2a: predicted pitch.

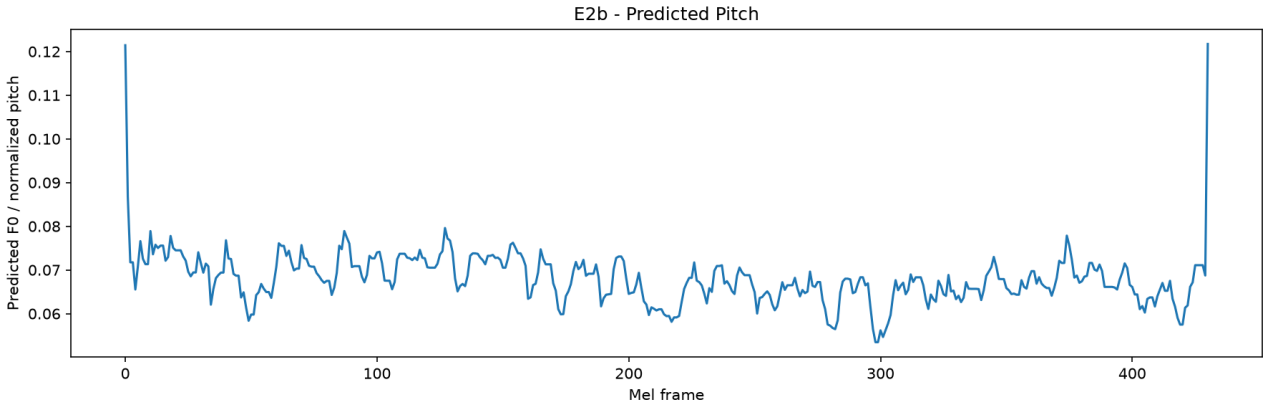


E2a: predicted energy.

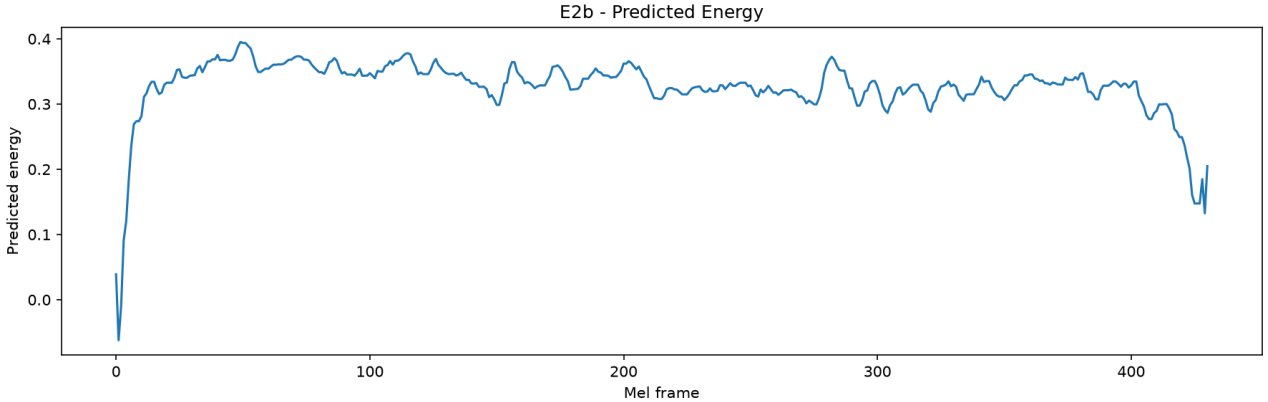
English - E2b



E2b: predicted duration.

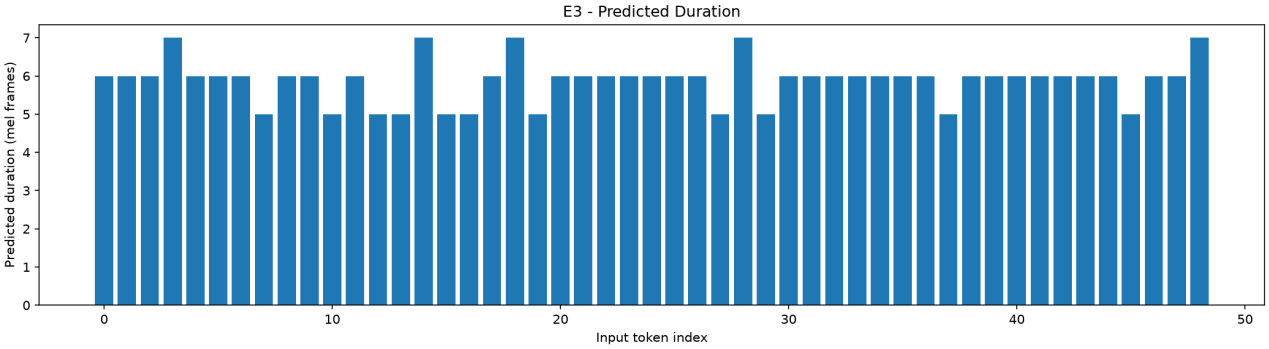


E2b: predicted pitch.

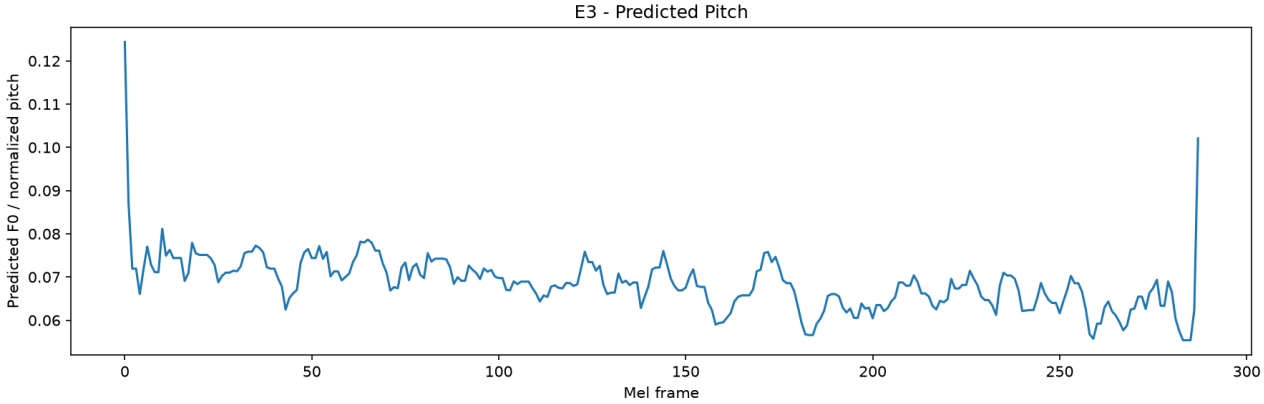


E2b: predicted energy.

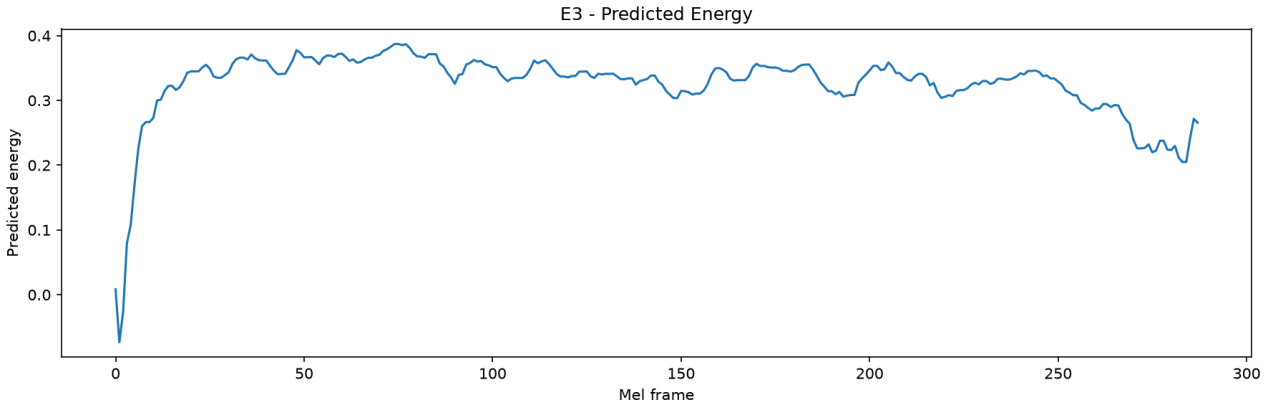
English - E3



E3: predicted duration.

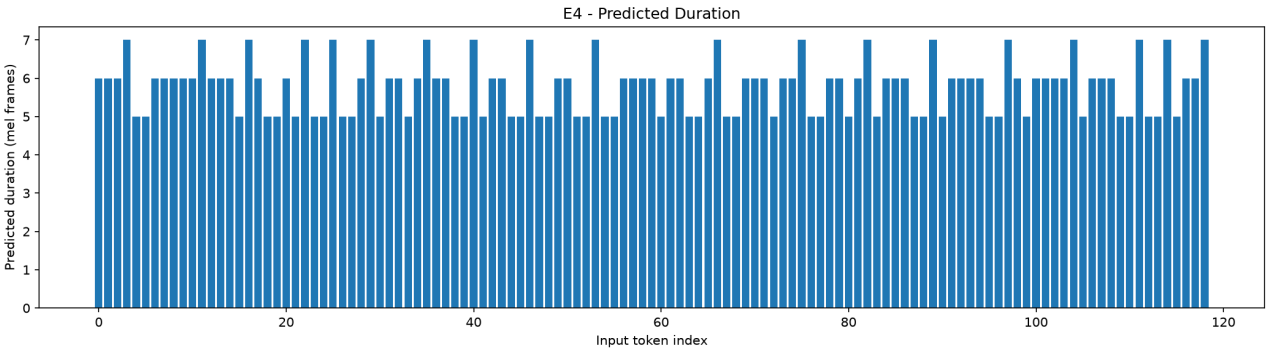


E3: predicted pitch.

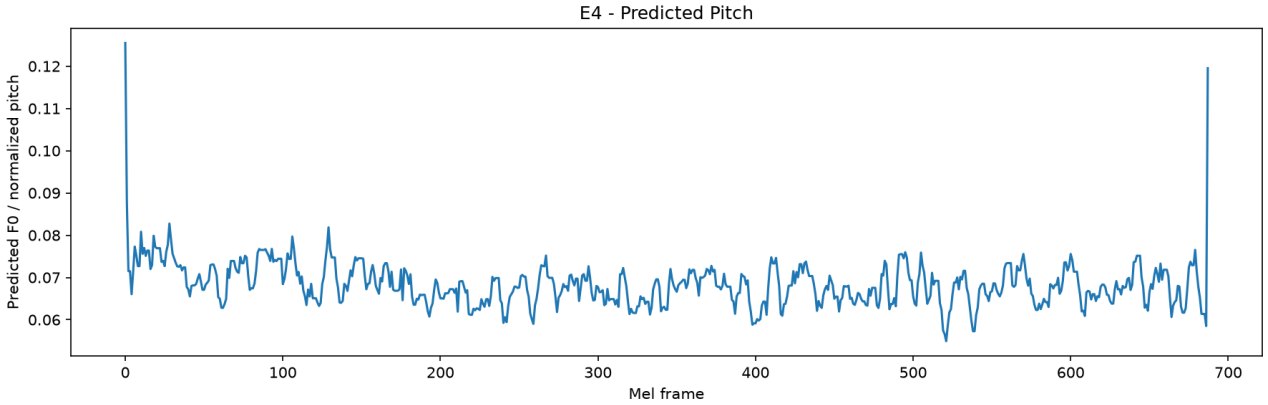


E3: predicted energy.

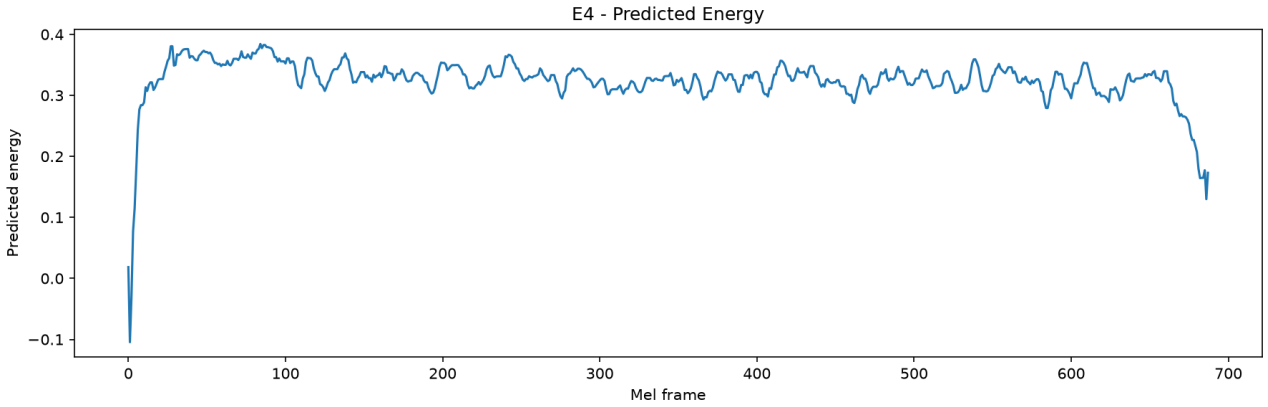
English - E4



E4: predicted duration.



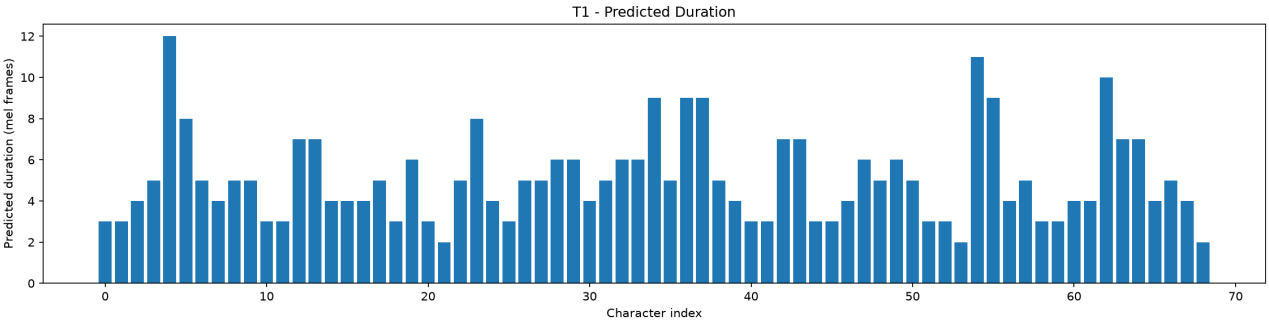
E4: predicted pitch.



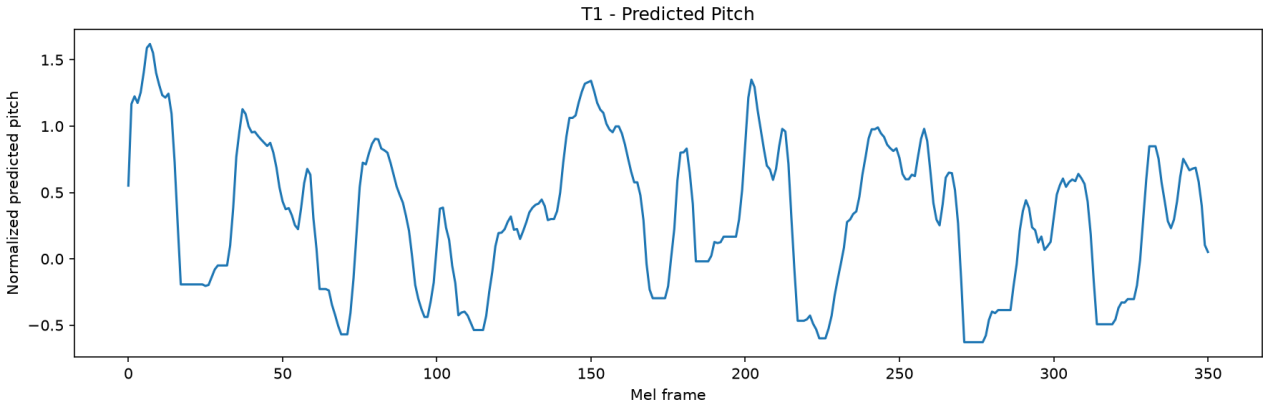
E4: predicted energy.

Telugu variance-adaptor plots

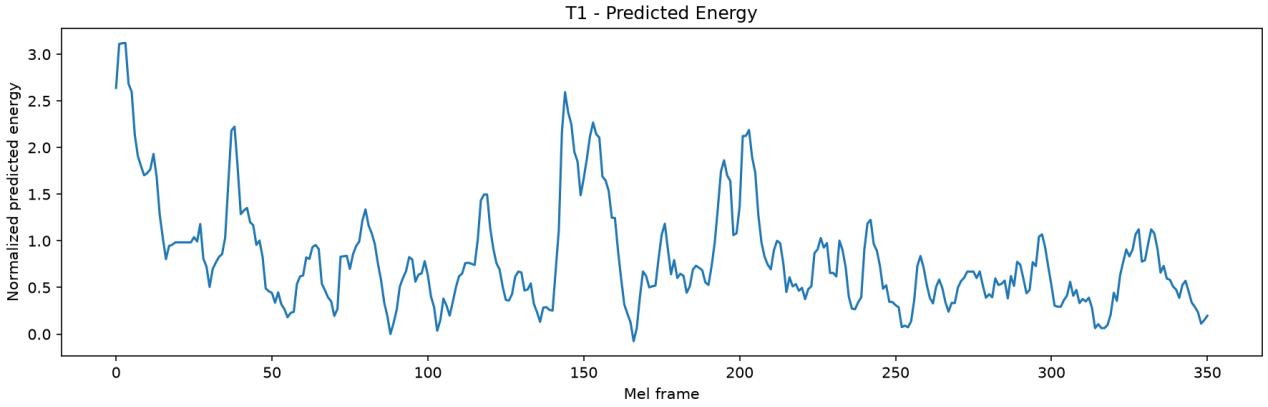
Telugu - T1



T1: predicted duration.

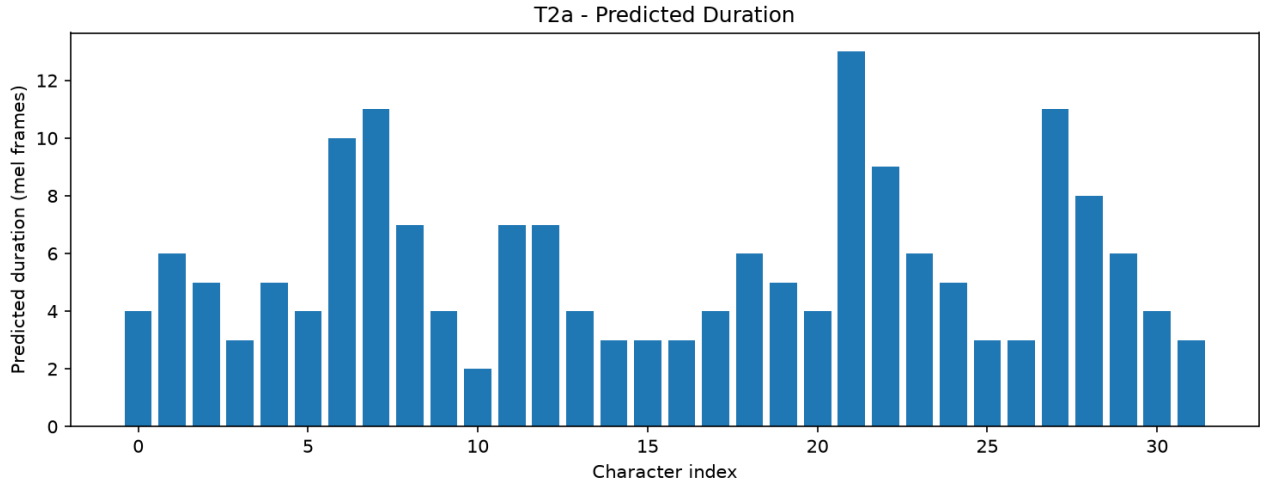


T1: predicted pitch.

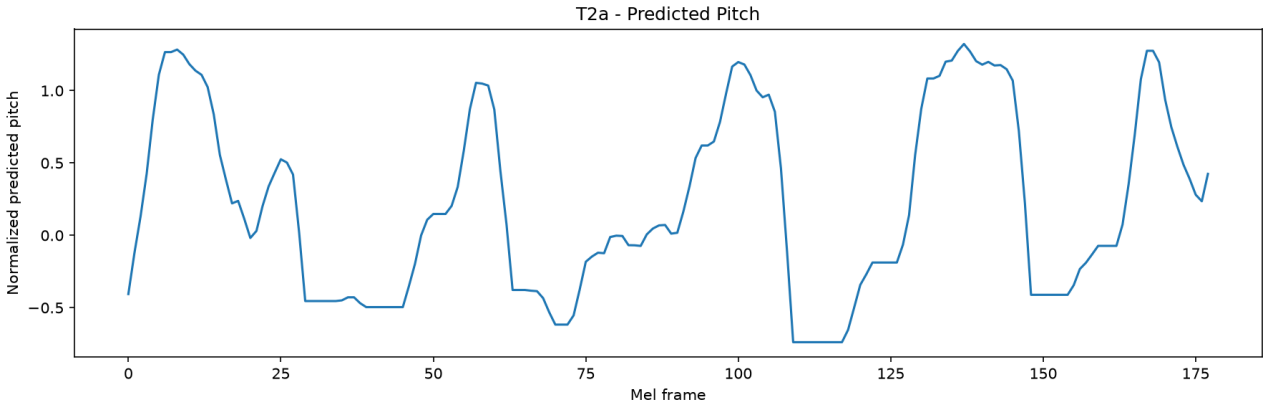


T1: predicted energy.

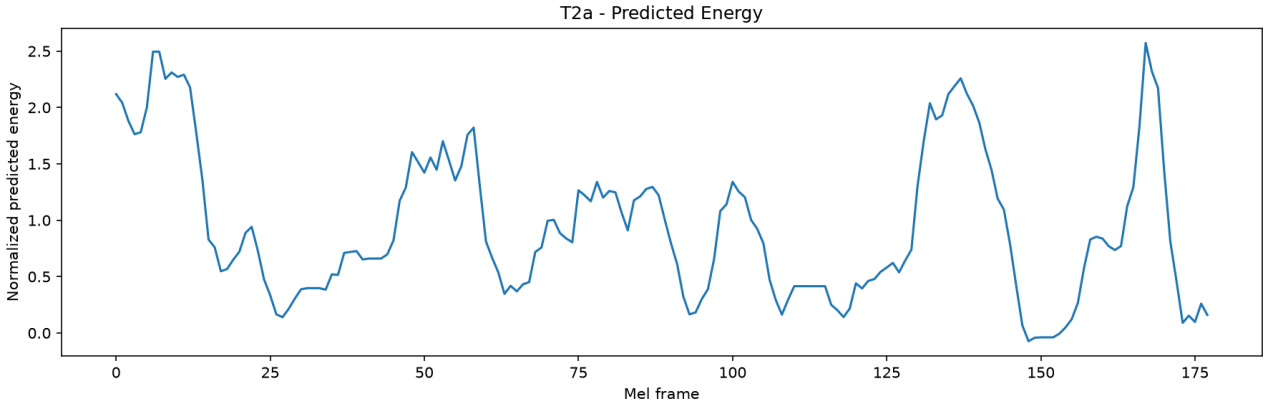
Telugu - T2a



T2a: predicted duration.

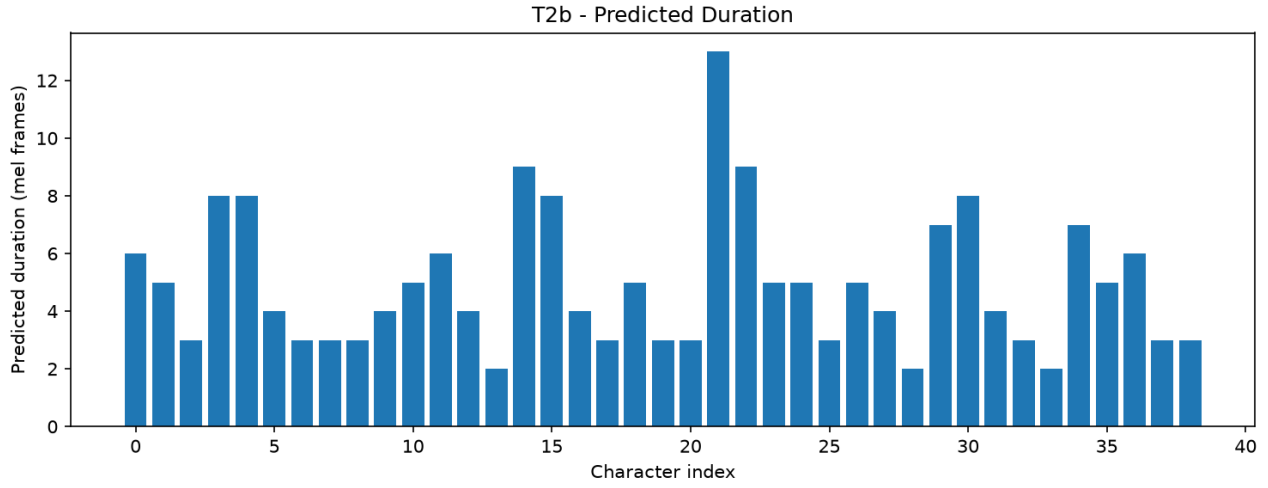


T2a: predicted pitch.

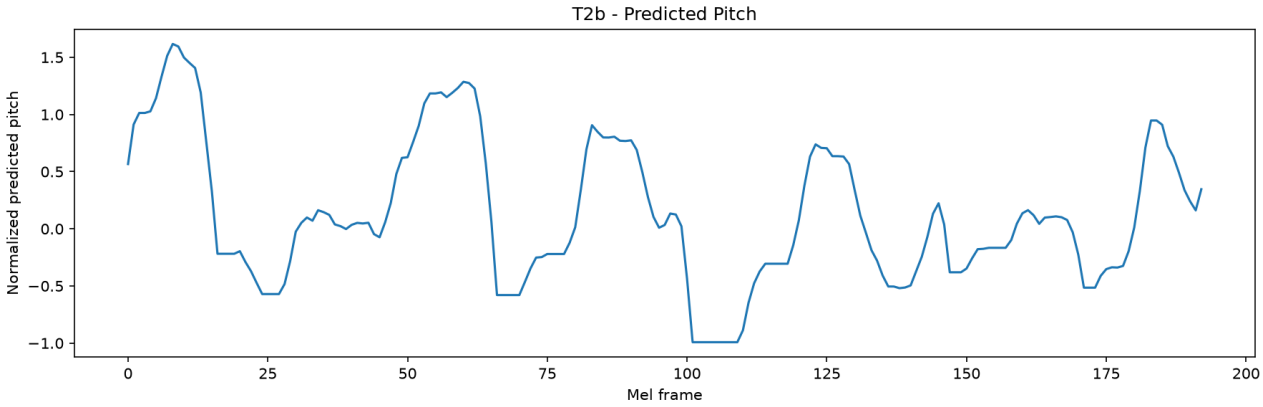


T2a: predicted energy.

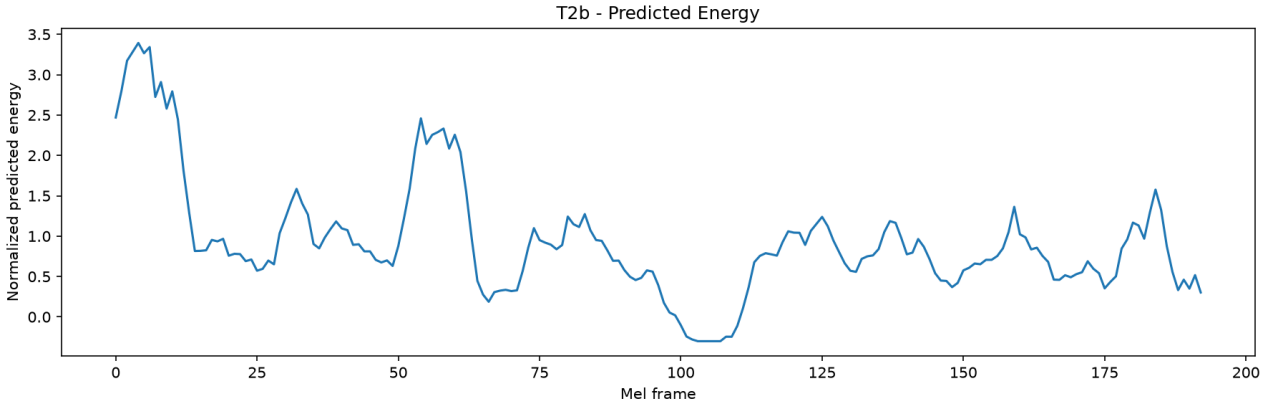
Telugu - T2b



T2b: predicted duration.

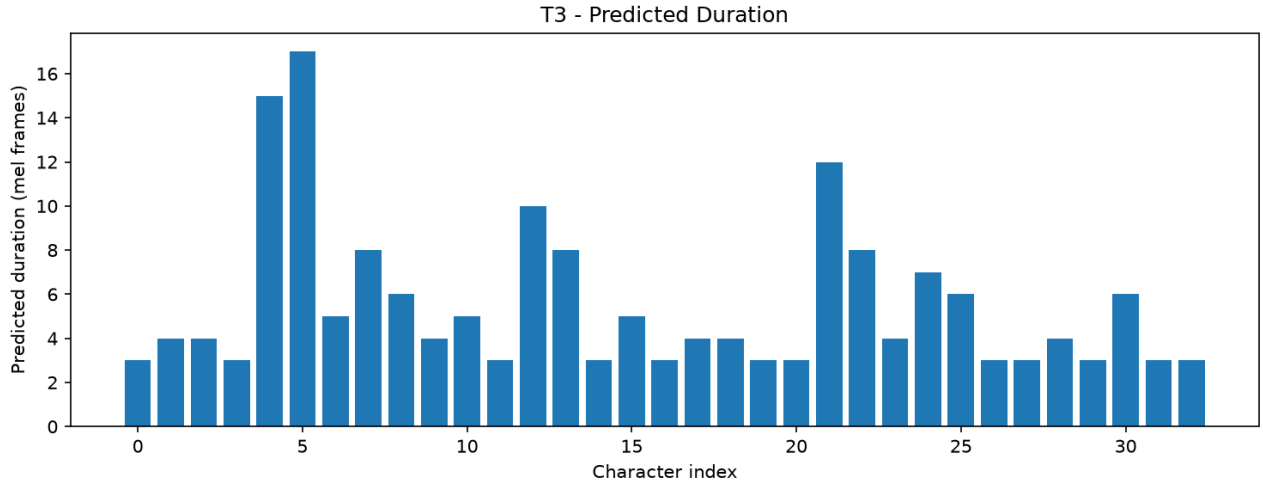


T2b: predicted pitch.

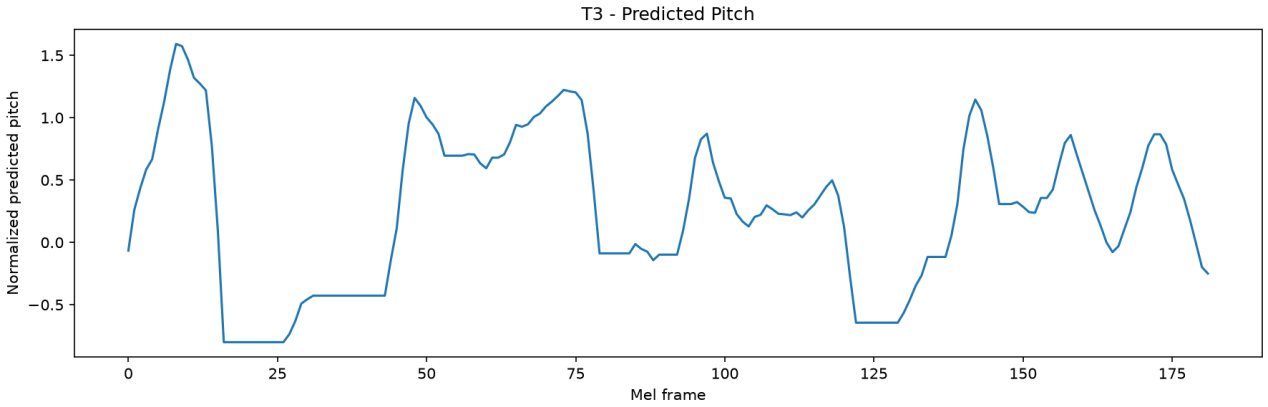


T2b: predicted energy.

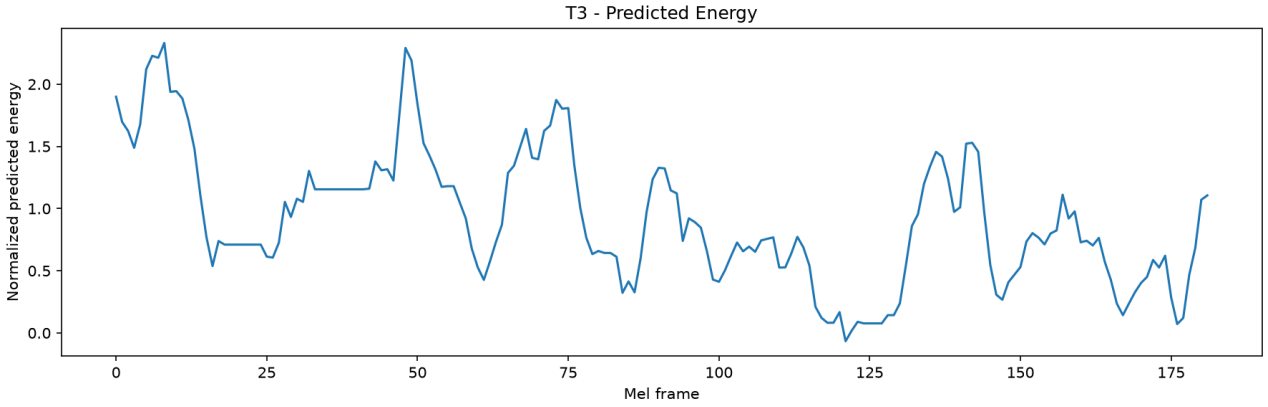
Telugu - T3



T3: predicted duration.

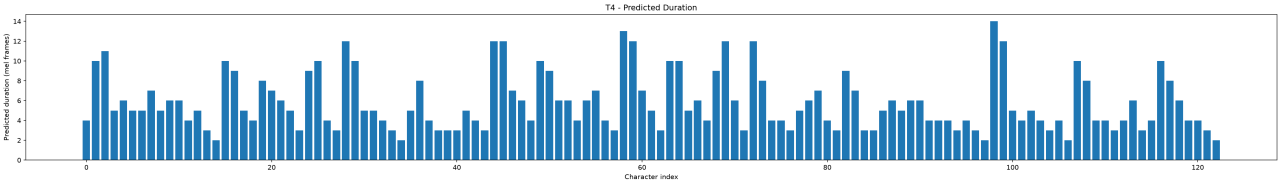


T3: predicted pitch.

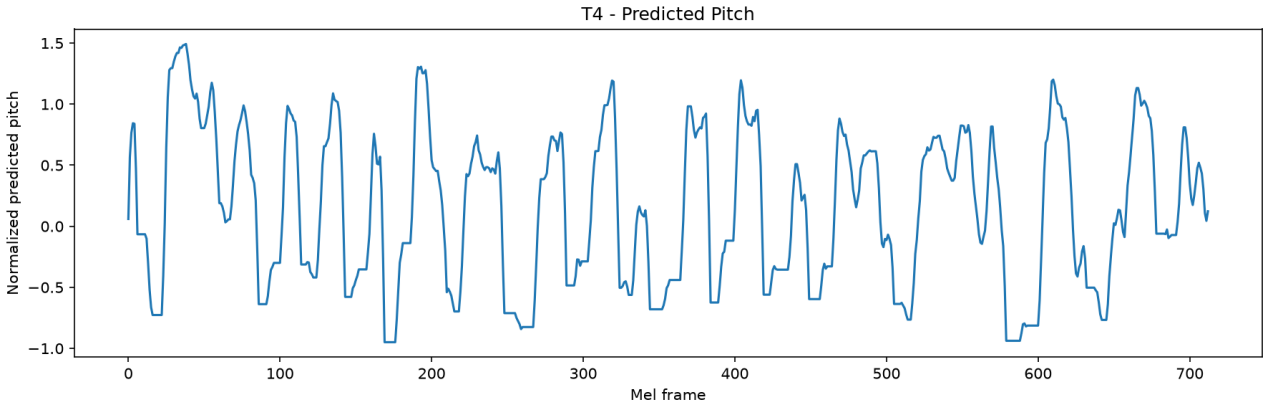


T3: predicted energy.

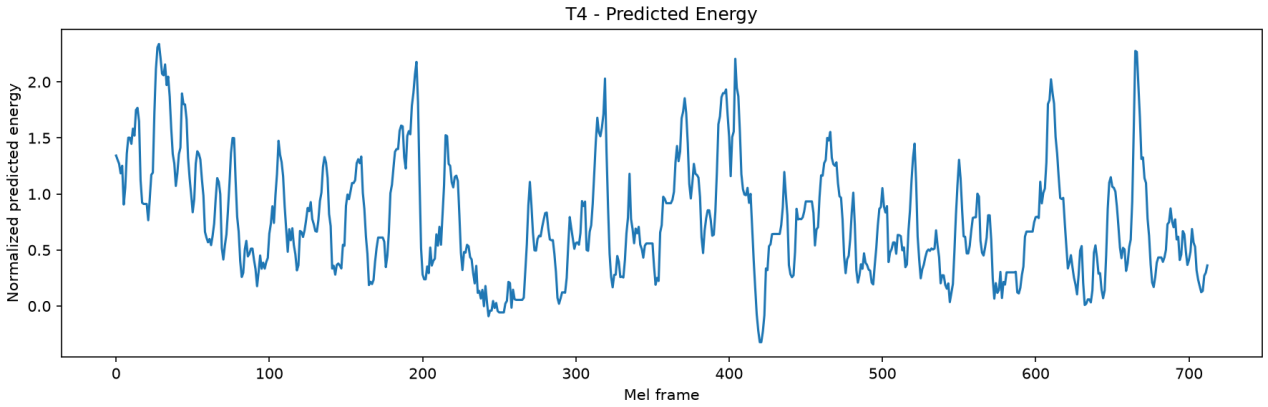
Telugu - T4



T4: predicted duration.



T4: predicted pitch.



T4: predicted energy.

8. Comparative Discussion

Aspect	English	Telugu
Input representation	Uppercase normalized English symbols	Telugu script and configured character vocabulary
Alignment	English wav2vec 2.0 CTC	Telugu wav2vec 2.0 CTC
Training examples	270	179
Validation examples	30	20
Completed epochs	1	1
Training loss	7.1523	8.5027
Validation loss	5.1646	5.2605
Waveform reconstruction	Griffin-Lim	Griffin-Lim

Both language pipelines completed feature preparation, one training epoch, checkpoint saving, and five-sentence inference. The loss values are not directly comparable measures of language quality because the datasets, token inventories, feature distributions, and training subsets differ.

9. Limitations

• Training was limited to one completed epoch per language because of the submission time constraint.

• Only a subset of the English aligned dataset was used for the reported training run.

• The Telugu dataset contained approximately 1.27 hours of usable aligned audio, below the originally requested duration.

• The models were trained on CPU with small batches and limited examples.

• Griffin-Lim reconstruction was used instead of a separately trained neural vocoder; this can reduce waveform naturalness and clarity.

• Telugu vocabulary and alignment constraints required text normalization for some test sentences.

• No formal human listening study, MOS evaluation, or objective intelligibility benchmark was conducted.

10. Reproducibility and Project Artifacts

The project contains separate English and Telugu model code/configuration directories, preprocessing scripts, training scripts, inference scripts, saved checkpoints, generated audio, plots, and metadata. Python 3.11 was used in a virtual environment. The experiments were executed on CPU.

Artifact	Location
English training	English/train_english.py
English inference	English/inference.py
English checkpoint	English/outputs/english/best_model.pth
Telugu training	Telugu/train_telugu.py
Telugu inference	Telugu/inference.py
Telugu checkpoint	Telugu/outputs/telugu/best_model.pth
English results	Results/English/
Telugu results	Results/Telugu/

The checkpoints and configuration files should be retained together. For exact reproduction, the corresponding datasets, preprocessing dependencies, aligner models, and model configurations are also required.

11. Conclusion

Separate FastSpeech 2 TTS pipelines for English and Telugu were implemented and exercised through training and inference. Both systems produced five test audio samples and the associated duration, pitch, and energy plots. Training and validation loss curves and model checkpoints were saved. The work demonstrates an end-to-end experimental implementation, while limited data preparation, one-epoch training, and Griffin-Lim reconstruction remain important limitations. Further training on the full target datasets and evaluation with a neural vocoder and listening tests would be necessary for a more robust quality assessment.